MADE on Statically Binarized MNIST
Reconstruction and Spatial Architectural Improvement

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Abstract

We reconstruct the one-hidden-layer, one-mask binarized-MNIST experiment from Masked Autoencoder for Distribution Estimation (MADE) [1], then replace its dense hidden transformation with a compact residual masked convolutional architecture. The reconstruction reaches 88.769 test nats, inside the paper's reported 88.40 ± 0.45 interval. The improved model reaches 87.399 nats with 653,281 parameters, reducing test NLL by 1.370 nats while using 94.8% fewer parameters. Under a matched unbiased-MMD protocol, sample quality does not improve, illustrating that likelihood and finite-sample similarity need not move together.

Part I
Paper Reconstruction

I.1 Selected paper condition

MADE turns a feed-forward autoencoder into an autoregressive density estimator by masking its connections. For a binary vector $x = (x_1, \dots, x_D)$,

$$p(x) = \prod_{d=1}^{D} p(x_d \mid x_{<d}). \tag{1}$$

All D conditionals are evaluated in one forward pass, while generation is sequential. We select statically binarized 28×28 MNIST, one fixed mask, and the paper target of 88.40 ± 0.45 test nats.

I.2 Architecture and masking

The reconstruction uses one 8,000-unit hidden layer with ReLU activation and Bernoulli outputs. The selected paper condition omits direct input-to-output conditioning weights. With masks M_W and M_V, the network is

$$h(x) = \mathrm{ReLU}(b + (W \odot M_W)x)\,, \tag{2}$$
$$a(x) = c + (V \odot M_V)h(x), \tag{3}$$
$$p(x_d = 1 \mid x_{<d}) = \sigma(a_d(x)). \tag{4}$$

Inputs and outputs receive degrees $1, \dots, D$; hidden degrees are sampled from $\{1, \dots, D-1\}$. Non-strict input–hidden and strict hidden–output comparisons prevent any output from accessing itself or a future value.

I.3 Objective and training set-up

For a binary example, the exact negative log-likelihood is

$$-\log p(x) = \sum_{d=1}^{D} \left[-x_d \log \sigma(a_d) - (1 - x_d) \log(1 - \sigma(a_d))\right]. \quad (5)$$

NLL in nats is the paper metric; lower values mean greater held-out probability and shorter ideal code length. Table 1 records the matched controls.

Table 1: Paper-faithful binarized-MNIST reconstruction set-up.

Item	Paper condition	Reconstruction
Data	static binary MNIST, 784 pixels	matched; 50,000 / 10,000 / 10,000
Architecture	one 8,000-unit ReLU hidden layer	matched
Connections	one fixed shuffled mask; no direct path	matched
Initialization	orthogonal weight matrices	matched
Objective	exact Bernoulli NLL	matched
Optimizer	Adagrad, $\eta = 0.01$, $\epsilon = 10^{-6}$, batch 100	matched
Stopping	validation NLL, patience 30	stopped after 77 epochs
Randomness	fixed experimental seed	seed 1234
Implementation	original Theano code	PyTorch, Lightning, Cyclopts

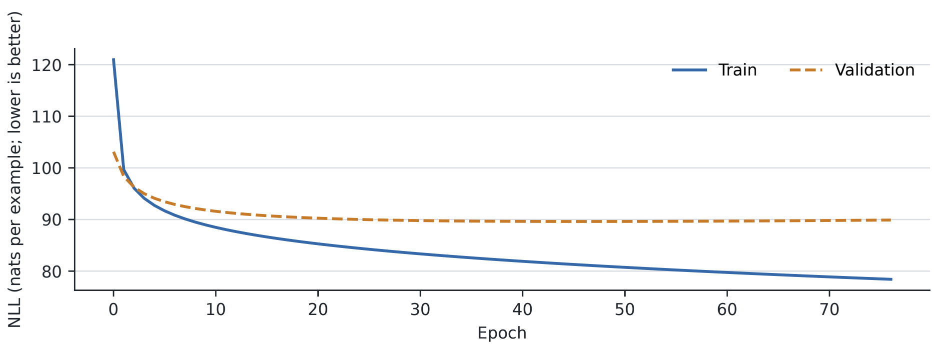

Figure 1: Reconstruction training and validation NLL. Validation NLL is minimized at epoch index 46 (89.613 nats); training stops after index 76.

I.4 Reconstruction result

The selected checkpoint reaches 88.769 test nats, an absolute difference of 0.369 nats (0.42%) from the paper's central result and inside its interval $[87.95, 88.85]$. Figure 2 shows ancestral samples.

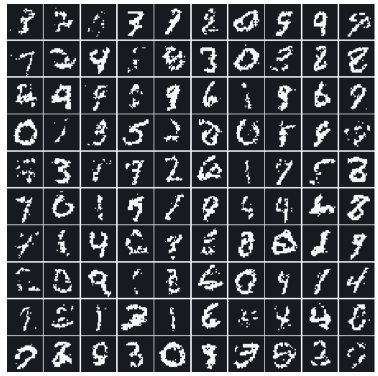

Figure 2: One hundred ancestral samples from reconstructed MADE (seed 1234).

Part II
Architectural Improvement

II.1 Spatial masked-convolution architecture

Dense MADE ignores the known 28×28 geometry. We therefore replace the dense hidden transformation with a compact PixelCNN-style decoder [2]: a strict type-A 7×7 masked input convolution, four type-B 3×3 masked residual blocks with 32 channels, and a 1×1 output projection. A strict masked linear path supplies global dependence on earlier pixels. The architecture uses a fixed raster ordering and 653,281 parameters. For residual block r,

$$h_0 = \mathrm{ReLU}(\mathrm{Conv}_A(x)), \tag{6}$$

$$h_{r+1} = \mathrm{ReLU}(h_r + \mathrm{Conv}_{B,r}(h_r)), \tag{7}$$

$$a(x) = \mathrm{Conv}_{1\times 1}(h_4) + (D \odot M_D)x. \tag{8}$$

Type A excludes the current pixel; type B includes the current hidden location but cannot introduce access to the current input. Automatic-Jacobian tests confirm zero derivative from every logit to itself and all future pixels. The dataset, Bernoulli NLL, optimizer, batch size, seed, and checkpoint rule remain unchanged.

II.2 Training and improved result

Figure 3 shows the complete 100-epoch trajectory. The best validation value occurs at the final checkpoint, where the full test NLL is 87.399 nats. This is 1.370 nats below the reconstruction and 1.001 nats below the paper's central value. It corresponds to 1.98 fewer ideal coding bits per image and about $e^{1.370} = 3.94$ times greater geometric-mean test probability.

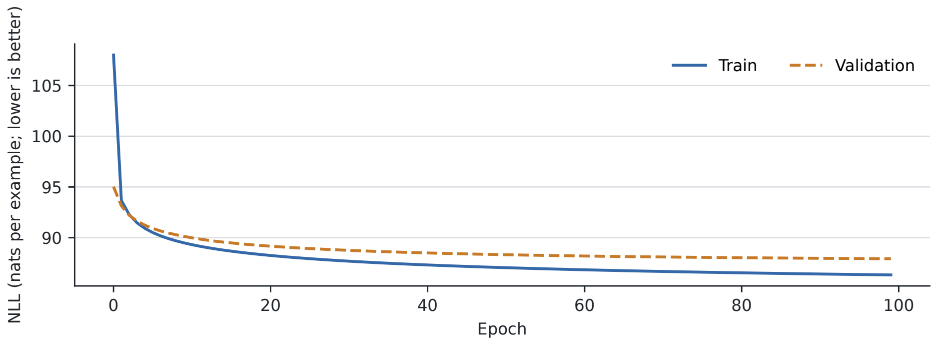

Figure 3: PixelCNN training and validation NLL across the initial five epochs and the resumed run through epoch index 99.

Figure 4: One hundred ancestral samples from the improved checkpoint (seed 1234).

Part III
Summary

III.1 Common evaluation standard

Every reconstructed checkpoint is selected by validation NLL and evaluated once on the complete 10,000-example test split. The course sample metric is unbiased squared MMD with the prespecified kernel $k(x, y) = \exp[-\|x - y\|_2^2/(2 \cdot 196)]$. Both models use the same first 100 held-out vectors and 100 ancestral samples generated with seed 1234. Runtime is end-to-end training wall time, and parameter counts include all trainable weights and biases.

III.2 Three-way comparison

Table 2: Paper, reconstruction, and improved-model results. Lower is better for NLL and MMD.

Metric	MADE paper	Reconstruction	Improved PixelCNN
Test NLL (nats)	88.40 ± 0.45	88.769	**87.399**
Unbiased MMD^2	—	**0.06497**	0.06836
Trainable parameters	12,552,784	12,552,784	**653,281**
Training wall time	not reported	~8m 31s	14m 53s
Selected epoch	not reported	47	100
Autoregressive ordering	fixed shuffled	fixed shuffled	fixed raster

III.3 Discussion

The spatial architecture produces a material likelihood gain while reducing parameter count by 94.8%, but native convolutional computation takes longer than the wide dense baseline. MMD^2 is slightly worse, so the likelihood gain does not imply a uniform sample-quality improvement under this finite-sample kernel test. A direct masked path improved NLL only marginally; degree-preserving residual and dense-bottleneck MADE variants did not justify full adoption. The successful result came from matching the architecture to image geometry while preserving exact autoregressive likelihood evaluation. One seed establishes the controlled result but not cross-seed statistical significance.

References

[1] Mathieu Germain, Karol Gregor, Iain Murray, and Hugo Larochelle. MADE: Masked autoencoder for distribution estimation. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 881–889. PMLR, 2015. URL https://proceedings.mlr.press/v37/germain15.html.

[2] A"aron van den Oord, Nal Kalchbrenner, Oriol Vinyals, Lasse Espeholt, Alex Graves, and Koray Kavukcuoglu. Conditional image generation with PixelCNN decoders. *arXiv preprint arXiv:1606.05328*, 2016. URL https://arxiv.org/abs/1606.05328.